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Composition Series for Primary Ideals

Notation 1.1. All rings will be unital and commutative.

Theorem 1.2

Let R be a ring. The following are equivalent:

- (1) R is Noetherian and every prime ideal of R is maximal.
- (2) R is Noetherian and $\dim R = 0$.
- (3) R is of finite length as an R -module.
- (4) R is Artinian.

Proof. Eisenbud Theorem 2.14. □

Corollary 1.3

Let R be a Noetherian ring. The following are equivalent:

- (1) Every prime ideal of R is maximal.
- (2) $\dim R = 0$.
- (3) R is of finite length as an R -module.
- (4) R is Artinian.

Proof. Follows from theorem 1.2 and the fact that Artinian rings are Noetherian. □

Corollary 1.4

Let R be a Noetherian ring and $\mathfrak{q}$ be a $\mathfrak{p}$ -primary ideal of R . Then $R/\mathfrak{q}$ is of finite length as an $R/\mathfrak{q}$ -module if and only if $\mathfrak{p}$ is a maximal ideal of R .

Moreover, any R -submodule of $R/\mathfrak{q}$ is a $R/\mathfrak{q}$ -submodule of $R/\mathfrak{q}$. So $R/\mathfrak{q}$ is of finite length as an $R/\mathfrak{q}$ -module if and only if $R/\mathfrak{q}$ is of finite length as an R -module.

Proof. R is Noetherian implies $R/\mathfrak{q}$ is Noetherian. By corollary 1.3, $R/\mathfrak{q}$ is of finite length as an $R/\mathfrak{q}$ -module if and only if every prime ideal of $R/\mathfrak{q}$ is maximal, which happens if and only if $\mathfrak{p}$ is a maximal ideal of R (because prime ideals of $R/\mathfrak{q}$ are in bijection with prime ideal of R containing $\mathfrak{q}$).

An R -submodule of $R/\mathfrak{q}$ is the quotient of a R -submodule of R containing the R -module $\mathfrak{q}$ by the R -module $\mathfrak{q}$. So it is of the form $I/\mathfrak{q}$ for some ideal $I \subseteq R$ containing $\mathfrak{q}$. Since $\mathfrak{q}$ annihilate $I/\mathfrak{q}$, it follows that $I/\mathfrak{q}$ can be viewed as a $R/\mathfrak{q}$. Conversely, any $R/\mathfrak{q}$ -submodule of $R/\mathfrak{q}$ can be viewed as an R -submodule of $R/\mathfrak{q}$ by the surjective ring homomorphism $R \rightarrow R/\mathfrak{q}$. □

Lemma 1.5

Let R be a ring. For any ideal $I \subseteq R$ and element $x \in R$, the following is a surjective R -module homomorphism:

$$\begin{aligned} R &\longrightarrow (I + (x))/I \\ 1 &\longmapsto x + I \\ a &\longmapsto ax + I \end{aligned}$$

with kernel $(I : x) = \{a \in R \mid ax \in I\}$. In particular, there is an R -module isomorphism $R/(I : x) \cong (I + (x))/I$.

Proof. The R -module homomorphism $R \rightarrow (I + (x))/I$ defined by $1 \mapsto x + I$ is surjective because $(I + (x))/I$ is generated by $x + I$ as an R -module (the coset of $(I + (x))/I$ represented by $i + ax \in I + (x)$ is the equal to the coset represented by ax which is the image of $a \in R$).

The kernel of this R -module homomorphism is by definition

$$\{a \in R \mid ax + I = 0 + I \text{ in } (I + (x))/I\} = \{a \in R \mid ax \in I\} = (I : x) \quad \square$$

Corollary 1.6

Let R be a ring. For any ideal $I \subseteq R$ and element $x \in R$, there is a bijection between the set of ideals $J \subseteq R$ such that $I \subseteq J \subseteq I + (x)$ and the set of ideals of R containing $(I : x)$. Moreover, the bijection is given by $J \mapsto I + xJ$.

Proof. This follows from the correspondence theorem of submodules applied to the isomorphism $R/(I : x) \cong (I + (x))/I$ in lemma 1.5. The isomorphism is given by $1 \mapsto x$, so an ideal $J/(I : x)$ is mapped to the submodule

$$\frac{I + xJ}{I} \subseteq \frac{I + (x)}{I} = \frac{I + xR}{I} \quad \square$$

Application 1.7. Let k be a field and $R = k[z_0, \dots, z_{c-1}]$ be a polynomial ring over k . Let $\mathfrak{m} = (x_0, \dots, x_{n-1})$ be a maximal ideal of R and $I = (f_0, \dots, f_{r-1})$ be an $\mathfrak{m}$ -primary ideal of R . The weird indexing is to match the indexing of the variables in Macaulay2.

The following is an algorithm to compute a maximal ascending chain of ideals $I = I_0 \subsetneq I_1 \subsetneq \dots \subsetneq I_s = \mathfrak{m}$ containing I up to the maximal ideal $\mathfrak{m}$. Taking the quotients of R by this chain gives a composition series of R/I .

- Define $M_i := I + (x_0, \dots, x_{i-1})$ for each $i \in \{0, \dots, n\}$.
- Define $Q_i := M_{i+1}/M_i$ for each $i \in \{0, \dots, n-1\}$.
- Define $J_i := (M_i : x_i)$ for each $i \in \{0, \dots, n-1\}$.

We begin the output chain with $I = M_0$ and for each $i \in \{0, \dots, n-1\}$, we attach a maximal ascending chain of ideal of R in between M_i and M_{i+1} (excluding M_i and including M_{i+1}) to the output chain. There are three cases:

Case 1. If $Q_i = 0$, then $M_{i+1} = M_i$. So we do not attach any ideals to the output chain.

Case 2. If Q_i is simple, then we attach M_{i+1} to the output chain.

Case 3. If Q_i is not simple, we invoke the isomorphism

$$R/J_i \cong Q_i = M_{i+1}/M_i$$

To get a maximal chain of ideals between M_i and M_{i+1} , we compute a maximal chain of ideals between J_i and R , and for each ideal K inside this chain, we attach $M_i + x_i K$ to the output chain (because this is the bijection in corollary 1.6).

We will not run into an infinite loop because R/J_i necessarily has smaller length than R/I as an R -module when $Q_i \neq 0$.

A Criterion for Ideal of Finite Colength

Theorem 2.1: Chinese Remainder Theorem

Let R be a ring and $I, J \subseteq R$ be ideals. If $I + J = R$, then the injective ring homomorphism

$$\begin{aligned} \varphi : R/(I \cap J) &\longrightarrow R/I \times R/J \\ a + (I \cap J) &\longmapsto (a + I, a + J) \end{aligned}$$

is also surjective. In particular, φ is an isomorphism

Proof. $I + J = R$ implies there exists $i \in I$ and $j \in J$ such that $1 = i + j$. Then

$$\varphi(i + (I \cap J)) = (0 + I, 1 + J) \quad \text{and} \quad \varphi(j + (I \cap J)) = (1 + I, 0 + J)$$

Thus for any $(a + I, b + J) \in R/I \times R/J$, we have

$$\varphi(bi + aj + (I \cap J)) = (a + I, b + J) \quad \square$$

Lemma 2.2

Let R be a ring and $I_1, \dots, I_n, J \subseteq R$ be ideals. Then

$$(I_1 \cap \dots \cap I_n) + J \subseteq (I_1 + J) \cap \dots \cap (I_n + J)$$

Moreover, if $I_1 + J = \dots = I_n + J = R$, the equality holds.

Proof. $(I_1 \cap \dots \cap I_n) + J \subseteq (I_1 + J) \cap \dots \cap (I_n + J)$ is clear. Suppose $I_1 + J = \dots = I_n + J = R$. Then for each $i \in \{1, \dots, n\}$, there exists $a_i \in I_i$ and $b_i \in J$ such that $1 = a_i + b_i$. It follows that

$$1 = (a_1 + b_1) \cdots (a_n + b_n) = (a_1 \cdots a_n) + \underbrace{\cdots}_{\text{all remaining term has a factor of some } b_i, \text{ so it is in } J} \in (I_1 \cap \dots \cap I_n) + J.$$

So $(I_1 \cap \dots \cap I_n) + J = R = (I_1 + J) \cap \dots \cap (I_n + J)$. $\square$

Corollary 2.3: Chinese Remainder Theorem

Let R be a ring and $I_1, \dots, I_n \subseteq R$ be ideals such that $I_i + I_j = R$ for all distinct $i, j \in \{1, \dots, n\}$. Then

$$\frac{R}{I_1 \cap \dots \cap I_n} \cong \frac{R}{I_1} \times \dots \times \frac{R}{I_n}$$

Proof. We have the chain of equalities by theorem 2.1:

$$\frac{R}{I_1 \cap \dots \cap I_n} \cong \frac{R}{I_1 \cap \dots \cap I_{n-1}} \times \frac{R}{I_n} \cong \dots \cong \frac{R}{I_1 \cap I_2} \times \frac{R}{I_3} \times \dots \times \frac{R}{I_n} \cong \frac{R}{I_1} \times \dots \times \frac{R}{I_n}$$

For the i -th isomorphism, the assumption we need to verify in order to invoke theorem 2.1 is that $(I_1 \cap \cdots \cap I_{i-1}) + I_i = R$. This follows from lemma 2.2 and the assumption that $I_j + I_i = R$ for all $j \in \{1, \dots, i-1\}$. $\square$

Lemma 2.4

Let R be a ring and $\mathfrak{q}_1$ be a $\mathfrak{p}_1$ -primary ideal and $\mathfrak{q}_2$ be a $\mathfrak{p}_2$ -primary ideal of R . If $\mathfrak{p}_1 + \mathfrak{p}_2 = R$, then $\mathfrak{q}_1 + \mathfrak{q}_2 = R$.

Proof. Suppose $\mathfrak{p}_1 + \mathfrak{p}_2 = R$, then there exists $a \in \mathfrak{p}_1$ and $b \in \mathfrak{p}_2$ such that $a + b = 1$. Since $\mathfrak{p}_1 = \sqrt{\mathfrak{q}_1}$ and $\mathfrak{p}_2 = \sqrt{\mathfrak{q}_2}$, it follows that there exists $m, n \in \mathbb{Z}_{>0}$ such that $a^m \in \mathfrak{q}_1$ and $b^n \in \mathfrak{q}_2$. Then

$$1 = (a + b)^{m+n} \in \mathfrak{q}_1 + \mathfrak{q}_2 \quad \square$$

Theorem 2.5

Let R be a noetherian ring and I be a proper ideal so that I admit a primary decomposition $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$. Let $\text{MSpec}(R) = \{\text{maximal ideals of } R\}$ and $\text{Ass}(I) = \{\mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_n\}$ where $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$.

Then R/I is of finite length as an R -module if and only if the $\text{Ass}(I) \subseteq \text{MSpec}(R)$.

Proof. Suppose $\text{Ass}(I) \subseteq \text{MSpec}(R)$. Then each $\mathfrak{p}_i$ is a maximal ideal and $\mathfrak{p}_i + \mathfrak{p}_j = R$ for distinct $i, j \in \{1, \dots, n\}$ since maximal ideals are pairwise comaximal.

Then by lemma 2.4, we have $\mathfrak{q}_i + \mathfrak{q}_j = R$ for distinct $i, j \in \{1, \dots, n\}$. By corollary 2.3, we have

$$\frac{R}{I} = \frac{R}{\mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n} \cong \frac{R}{\mathfrak{q}_1} \times \cdots \times \frac{R}{\mathfrak{q}_n}$$

By corollary 1.4, each $R/\mathfrak{q}_i$ is of finite length as R -modules because $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$ is a maximal ideal. Product of finite length modules is of finite length, so R/I is of finite length as an R -module.

Conversely, suppose $\text{Ass}(I) \not\subseteq \text{MSpec}(R)$. Then there exists $i \in \{1, \dots, n\}$ such that $\mathfrak{p}_i$ is not a maximal ideal. Now $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n \subseteq \mathfrak{q}_i \subseteq \mathfrak{p}_i$ which implies there exists a surjective ring homomorphism $R/I \rightarrow R/\mathfrak{p}_i$. Since $\mathfrak{p}_i$ is a prime ideal but not a maximal ideal, It follows that $\dim R/\mathfrak{p}_i \geq 1$. Then $\dim R/I \geq 1$ as well which implies R/I is not of finite length as an R/I -module by corollary 1.3. By corollary 1.4, R/I is not of finite length as an R -module. $\square$